Chapter 9 solutions

CH09-E1

1.

$$(2-i)(3+2i) = 6 + 4i - 3i - 2i^2 = 8 + i.$$

$$\frac{2-i}{3+2i} = \frac{(2-i)(3-2i)}{13} = \frac{4-7i}{13}.$$

2.

$$|1 + i\sqrt{3}| = 2, \qquad 1 + i\sqrt{3} = 2e^{i\pi/3}.$$

3.

$$(1 + i\sqrt{3})^6 = 2^6 e^{i6\pi/3} = 64e^{2\pi i} = 64.$$

4.

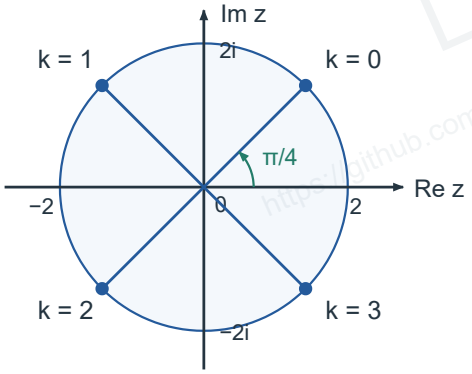
$$z^4 = -16 = 16e^{i\pi}.$$

Roots:

$$z_k = 2e^{i(\pi + 2\pi k)/4}, \quad k = 0, 1, 2, 3.$$

The arguments are $\pi/4, 3\pi/4, 5\pi/4, 7\pi/4$, equally spaced on the radius-2 circle.

z⁴ = −16: radius 2, angle spacing π/2



Angles counterclockwise from the positive real axis

k	$\theta_k = (\pi + 2\pi k)/4$	z_k
0	π/4	√2 + i√2
1	3π/4	−√2 + i√2
2	5π/4	−√2 − i√2
3	7π/4	√2 − i√2

$|z_k| = 2$; $4\theta_k = \pi + 2\pi k$, hence $z_k^4 = -16$

CH09-E2

1.

$$\|u\|^2 = |1+i|^2 + |2-i|^2 = 7,$$

$$\|v\|^2 = 7.$$

$$u^H v = (1-i)(2-i) + (2+i)(1+i).$$

The two terms are $1 - 3i$ and $1 + 3i$, so

$$u^H v = 2.$$

$$v^H u = 2.$$

Chapter 9 solutions (continued)

2.

$$u^H v = \overline{v^H u} = \overline{2} = 2.$$

3. The inner product is 2, not zero, so the vectors are not orthogonal.

4.

$$q = \frac{1}{\sqrt{7}}(1+i, 2-i)^T,$$
$$q^H q = \frac{1}{7} u^H u = 1.$$

CH09-E3

1.

$$S_{12} = i, \qquad S_{21} = -i = \overline{i},$$

Both diagonal entries are real, and the off-diagonal entries are conjugates, proving $S^H = S$.

2.

$$p_S(\lambda) = (2-\lambda)^2 - 1 = (\lambda - 1)(\lambda - 3).$$

Both eigenvalues, 1 and 3, are real.

3. For $\lambda = 3$,

$$-z_1 + iz_2 = 0 \Rightarrow z_1 = iz_2.$$
$$q_1 = \frac{1}{\sqrt{2}}(i, 1)^T.$$

For $\lambda = 1$,

$$z_1 + iz_2 = 0 \Rightarrow z_1 = -iz_2.$$
$$q_2 = \frac{1}{\sqrt{2}}(-i, 1)^T.$$

4.

$$Q = \frac{1}{\sqrt{2}} \begin{bmatrix} i & -i \\ 1 & 1 \end{bmatrix}.$$
$$Q^H Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Therefore

$$S = Q \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} Q^H.$$

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Chapter 9 solutions (continued)

CH09-E4

1.

$$(A^H)^H = A.$$

For a product,

$$(AB)^H_{ij} = \overline{\sum_k A_{jk}B_{ki}} = \sum_k \overline{B_{ki}A_{jk}} = (B^H A^H)_{ij}.$$

2. If $A^H = -A$ and $Au = \lambda u$, with $u \neq 0$,

$$\lambda u^H u = u^H Au = -u^H A^H u = -(Au)^H u = -\overline{\lambda} u^H u.$$

Thus $\lambda = -\overline{\lambda}$, so its real part is zero.

3.

$$1 = \det(Q^H Q) = |\det Q|^2,$$

Hence the determinant modulus is 1.

4. Hermitian positive definite:

$$z^H Az = \lambda z^H z > 0$$

For a nonzero eigenvector this forces $\lambda > 0$.

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CH09-E5

1. The given matrix is

$$Q = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}.$$

$$Q^T Q = \frac{1}{2} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = I.$$

2. The standard basis vectors have inner product zero. Unitarity gives

$$(Qu)^H (Qv) = u^H Q^H Qv = 0.$$

Their norms also remain 1.

3.

$$\det(Q - \lambda I) = \lambda^2 - \sqrt{2}\lambda + 1.$$

$$\lambda = e^{\pm i\pi/4}, \qquad |\lambda| = 1.$$

4. The real matrix has $Q^T = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \neq Q$, so it is not Hermitian. Its nonreal eigenvalues give another check. Some matrices, such as the identity, are both unitary and Hermitian; neither property generally implies the other.

Chapter 9 solutions (continued)

CH09-E6

1. $\omega = i$:

$$F_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{bmatrix}.$$

2. $c = (1, 2, 0, -1)^T$:

$$y_0 = 2,$$
$$y_1 = 1 + 2i + 0 + i = 1 + 3i,$$
$$y_2 = 1 - 2 + 0 + 1 = 0,$$
$$y_3 = 1 - 2i + 0 - i = 1 - 3i.$$

Therefore

$$y = (2, 1 + 3i, 0, 1 - 3i)^T.$$

3.

$$F_4^H = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{bmatrix}.$$
$$F_4^H y = (4, 8, 0, -4)^T,$$

Division by 4 recovers $c = (1, 2, 0, -1)^T$.

4.

$$\|c\|^2 = 1 + 4 + 0 + 1 = 6.$$
$$\|y\|^2 = 4 + 10 + 0 + 10 = 24 = 4\|c\|^2.$$

CH09-E7

1. The even-indexed coefficients are

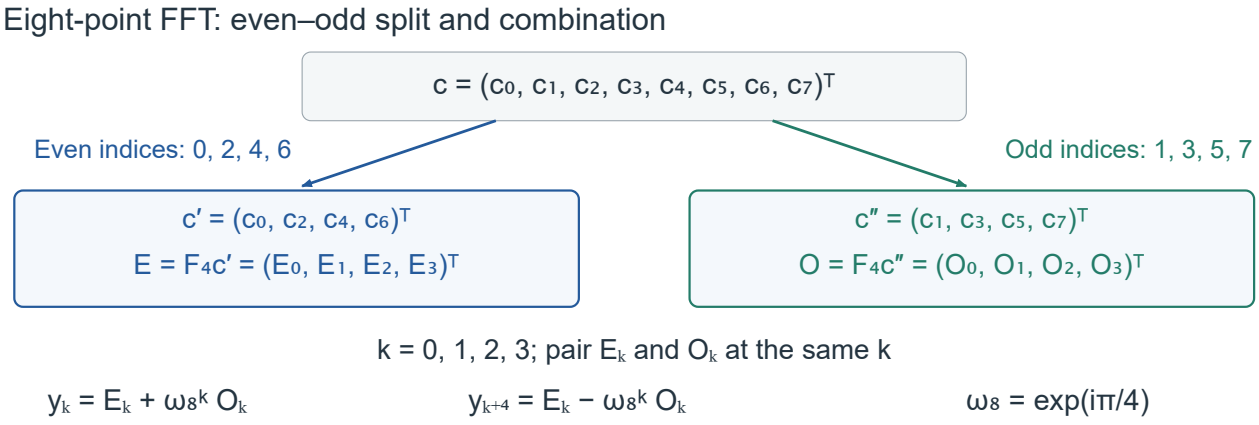
$$c' = (c_0, c_2, c_4, c_6)^T,$$

The odd-indexed coefficients are

$$c'' = (c_1, c_3, c_5, c_7)^T.$$

Taking their size-4 DFTs gives $E = F_4 c'$, $O = F_4 c''$.

Chapter 9 solutions (continued)



2. $j = 0, \ldots, 3$:

$$y_j = E_j + \omega_8^j O_j,$$
$$y_{j+4} = E_j - \omega_8^j O_j.$$

The twiddle factor is ω_8^j.

3.

$$T(8) = 2T(4) + 8,$$
$$T(4) = 2T(2) + 4, \qquad T(2) = 2T(1) + 2.$$

Each stage costs linear work and there are $\log_2 n$ stages, so

$$T(n) = \Theta(n \log n).$$

4. Radix-2 requires a power-of-two length. Other lengths can use mixed radices, Bluestein, or zero-padding; padding changes the signal length and frequency grid.

CH09-E8

1. Let q_j be column j. From $Q^H Q = I$,

$$q_i^H q_j = \delta_{ij},$$

these n columns form an orthonormal basis of $\mathbb{C}^n$. For square Q, $Q^H Q = I$ also gives $Q^{-1} = Q^H$, hence $QQ^H = I$. Writing row i as the column $r_i = (Q_{i1}, \ldots, Q_{in})^T$,

$$r_i^H r_j = \overline{(QQ^H)_{ij}} = \delta_{ij}.$$

the rows also form an orthonormal basis. The two matrix products directly test different vector families.

2. A Hermitian matrix $S = S^H$ has real diagonal entries and hence real trace. Its determinant is the product of real eigenvalues and is real as well.

3.

Chapter 9 solutions (continued)

$$(cF_n)^H(cF_n) = |c|^2 nI.$$

A unitary matrix satisfies

$$|c| = \frac{1}{\sqrt{n}}.$$

> Any global phase $e^{i\phi}$ preserves the norm.

4. Choosing ω or $\overline{\omega}$ exchanges the forward/inverse convention, or corresponds to conjugation and reversed order. After normalization,
$$F_n F_n^H = nI.$$

the factor n in Parseval comes from normalization, not the sign convention.

CH09-E9

1. Bit reversal reorders $0, 1, 2, 3$ as $0, 2, 1, 3$, giving
$$(c_0, c_2, c_1, c_3) = (1, 0, 2, -1).$$
2. The first stage is
$$(1, 0) \mapsto (1, 1), \qquad (2, -1) \mapsto (1, 3).$$

The second stage is

$$(1, 1) \mapsto (2, 0), \qquad (1, 3) \mapsto (1 + 3i, 1 - 3i).$$

3. Therefore
$$F_4 c = (2, 1 + 3i, 0, 1 - 3i)^T.$$
4. Direct evaluation gives
$$y_1 = 1 + 2i + 0(-1) + (-1)(-i) = 1 + 3i,$$
$$y_3 = 1 - 2i + 0(-1) + (-1)i = 1 - 3i,$$

agreeing with the FFT stages.

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Chapter 9 solutions (continued)

Verification code appendix: complex_fft.py

```
"""Verify Hermitian, unitary, DFT and radix-2 FFT examples."""

from cmath import exp, isclose, pi

Complex = complex
Vec = tuple[Complex, ...]
Mat = tuple[tuple[Complex, ...], ...]


def conjugate_transpose(a: Mat) -> Mat:
    return tuple(tuple(a[j][i].conjugate() for j in range(len(a))) for i in range(len(a[0])))


def matvec(a: Mat, x: Vec) -> Vec:
    return tuple(sum(a[i][j] * x[j] for j in range(len(x))) for i in range(len(a)))


def matmul(a: Mat, b: Mat) -> Mat:
    return tuple(
        tuple(sum(a[i][k] * b[k][j] for k in range(len(b))) for j in range(len(b[0])))
        for i in range(len(a))
    )


def matrix_power(a: Mat, exponent: int) -> Mat:
    result = tuple(tuple(1.0 + 0j if i == j else 0j for j in range(len(a))) for i in
range(len(a)))
    while exponent:
        if exponent & 1:
            result = matmul(result, a)
        a = matmul(a, a)
        exponent >>= 1
    return result


a = ((2 + 0j, 1j), (-1j, 3 + 0j))
assert conjugate_transpose(a) == a
```

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Chapter 9 solutions (continued)

```
traces = sum(a[i][i] for i in range(2))
assert abs(traces.imag) < 1e-12

q = (
    (1 / 2**0.5, 1 / 2**0.5),
    (-1 / 2**0.5, 1 / 2**0.5),
)
qh = conjugate_transpose(q)
qhq = matmul(qh, q)
assert qh != q
assert all(
    abs(qhq[i][j] - (1 if i == j else 0)) < 1e-12
    for i in range(2)
    for j in range(2)
)


def naive_dft(values: Vec) -> Vec:
    n = len(values)
    omega = exp(2j * pi / n)
    return tuple(
        sum((omega ** (j * k)) * values[k] for k in range(n))
        for j in range(n)
    )


def fft(values: Vec) -> Vec:
    n = len(values)
    if n == 1:
        return values
    if n & (n - 1):
        raise ValueError("power-of-two length required")
    even = fft(values[0::2])
    odd = fft(values[1::2])
    omega = exp(2j * pi / n)
```

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Chapter 9 solutions (continued)

```
    output = [0j] * n
    for j in range(n // 2):
        twiddle = (omega**j) * odd[j]
        output[j] = even[j] + twiddle
        output[j + n // 2] = even[j] - twiddle
    return tuple(output)


samples = (1 + 0j, 2 + 0j, 0 + 0j, -1 + 0j, 3 + 0j, -2 + 0j, 1 + 0j, 0.5 + 0j)
naive = naive_dft(samples)
fast = fft(samples)
assert all(isclose(naive[i], fast[i], abs_tol=1e-12) for i in range(8))

s4 = samples[:4]
y4 = naive_dft(s4)
assert abs(sum(abs(value) ** 2 for value in y4) - 4 * sum(abs(value) ** 2 for value in s4)) <
1e-10

signal = (1.0 + 0j, 2.0 + 0j, 0.0 + 0j, -1.0 + 0j)
reversed_signal = (signal[0], signal[2], signal[1], signal[3])
assert reversed_signal == (1.0 + 0j, 0.0 + 0j, 2.0 + 0j, -1.0 + 0j)
stage1 = (
    reversed_signal[0] + reversed_signal[1],
    reversed_signal[0] - reversed_signal[1],
    reversed_signal[2] + reversed_signal[3],
    reversed_signal[2] - reversed_signal[3],
)
assert stage1 == (1.0 + 0j, 1.0 + 0j, 1.0 + 0j, 3.0 + 0j)
stage2 = (
    stage1[0] + stage1[2],
    stage1[1] + 1j * stage1[3],
    stage1[0] - stage1[2],
    stage1[1] - 1j * stage1[3],
)
expected_trace = (2.0 + 0j, 1.0 + 3j, 0.0 + 0j, 1.0 - 3j)
assert all(isclose(stage2[i], expected_trace[i], abs_tol=1e-12) for i in range(4))
assert all(isclose(naive_dft(signal)[i], expected_trace[i], abs_tol=1e-12) for i in range(4))
```

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Chapter 9 solutions (continued)

```
def fft4_trace(signal):
    """Positive-sign DFT: bit reversal, adjacent pairs, then pairs (0,2)/(1,3)."""
    if len(signal) != 4:
        raise ValueError("exactly four coefficients required")
    reordered = (signal[0], signal[2], signal[1], signal[3])
    first = (
        reordered[0] + reordered[1], reordered[0] - reordered[1],
        reordered[2] + reordered[3], reordered[2] - reordered[3],
    )
    second = (
        first[0] + first[2], first[1] + 1j * first[3],
        first[0] - first[2], first[1] - 1j * first[3],
    )
    return reordered, first, second


counterexample = (2, 5, 7, -1)
reordered, first, second = fft4_trace(counterexample)
assert reordered == (2, 7, 5, -1) and first == (9, -5, 4, 6)
assert second == (13, -5 + 6j, 5, -5 - 6j)
assert all(isclose(a, b, abs_tol=1e-12) for a, b in zip(second, naive_dft(counterexample)))
from itertools import product
for coefficients in product(range(-2, 3), repeat=4):
    traced = fft4_trace(coefficients)[2]
    assert all(isclose(a, b, abs_tol=1e-12) for a, b in zip(traced, naive_dft(coefficients)))

print("Hermitian matrix trace imaginary part =", traces.imag)
print("Q^H Q =", qhq)
print("DFT and FFT match:", all(isclose(naive[i], fast[i], abs_tol=1e-12) for i in range(8)))
print("Parseval check passed")
print("n=4 FFT stages =", stage1, stage2)
```

References and attribution

Gilbert Strang. Introduction to Linear Algebra, 5th Edition, 2016.

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